



Writing small numbers in standard form

Task A: Writing very small numbers in standard form

1) Using the place value charts, write the following in standard form.

a)

O	t	h	th	tth	hth
10^0	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}
0	● 0	0	0	6	9

$$6.9 \times 10^{-4}$$

b)

O	t	h	th	tth	hth
10^0	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}
0	● 0	1	2	8	

$$1.28 \times 10^{-2}$$

2) Which of the following are not in standard form? Explain why.

a) 123×10^{-9} This is incorrect as 123 needs to be a number satisfying $1 \leq A < 10$

b) 1.24×10^{-3} This is in correct standard form

c) 0.96×10^{-1} This is incorrect as 0.96 needs to be a number satisfying $1 \leq A < 10$

d) $12\,300 \div 10^{10}$ This is incorrect as standard form uses multiplication.

e) 0.059×10^8 This is incorrect as 0.059 needs to be a number satisfying $1 \leq A < 10$

3) Without a place value chart, write the following in standard form.

a) $0.000\,000\,354 = 3.54 \times 10^{-7}$

b) $0.001\,45 = 1.45 \times 10^{-3}$

c) $0.000\,000\,000\,563 = 5.63 \times 10^{-10}$

d) $0.048\,6 = 4.86 \times 10^{-2}$

e) $0.000\,000\,000\,631 = 6.31 \times 10^{-10}$



Task B: Converting between ordinary and standard form

1) Convert the following into standard form.

- a) $0.000\ 000\ 263 = 2.63 \times 10^{-7}$
 b) $652\ 000\ 000 = 6.52 \times 10^8$
 c) $0.005\ 63 = 5.63 \times 10^{-3}$
 d) $892\ 000\ 000\ 000 = 8.92 \times 10^{11}$
 e) $0.000\ 000\ 000\ 000\ 0003 = 3 \times 10^{-16}$
 f) $30\ 004\ 000 = 3.0004 \times 10^7$
 g) $0.000\ 300\ 4 = 3.004 \times 10^{-4}$

2) Convert the following into ordinary numbers.

- a) $2.31 \times 10^4 = 23\ 100$
 b) $6.153 \times 10^{-5} = 0.000\ 061\ 53$
 c) $9.1565 \times 10^{-6} = 0.000\ 009\ 156\ 5$
 d) $2.48 \times 10^5 = 248\ 000$
 e) $2.48 \times 10^{-5} = 0.000\ 024\ 8$
 f) $3.0065 \times 10^{-9} = 0.000\ 000\ 003\ 006\ 5$

 3) A microchip has a width of 2×10^{-6} m. Write this as an ordinary number in cm.

$$0.000\ 002\ m = 0.000\ 000\ 02\ cm$$

4) Mars is approximately 225 million km away from Earth. Write this in standard form in m.

$$225\ 000\ 000\ km = 225\ 000\ 000\ 000\ m = 2.25 \times 10^{11}\ m$$

 5) The radius of an atom is approximately 1×10^{-10} m. Write this as an ordinary number in mm.

$$0.000\ 000\ 000\ 1\ m = 0.000\ 000\ 000\ 000\ 1\ mm$$



Task C: Standard form on the calculator

1) Write the output as a decimal.



0.000 068



0.000 000 298



0.009 378



0.000 69

2) Investigate when the number becomes so small that the calculator converts it into standard form.

Any number when written in standard form has an exponent less than or equal to -3

3) Fill in the missing information.

